

MUHAMMAD HAIDER ZAIDI

☎ +49 1523 1698988 ✉ mhaiderzaidi21@gmail.com in [linkedin.com/in/muhammad-haider-zaidi](https://www.linkedin.com/in/muhammad-haider-zaidi)
🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Bash, JavaScript

Tools/Frameworks: Google Cloud Platform (GCP), Vertex AI, BigQuery, Cloud Storage, Cloud Run, Cloud Functions, GKE, Terraform, dbt, Docker, Kubernetes, TensorFlow, PyTorch, FastAPI, Flask, TensorRT, YOLO, Airflow, PySpark

Competence: Machine Learning, MLOps, CI/CD Pipelines, Data Engineering, Model Deployment, AI Systems Monitoring, Software Architecture, Clean Code Principles, Data Visualization, Feature Engineering

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Developed physics-informed hybrid loss functions and novel architectures, enhancing model accuracy and stability.
- Collaborated with PhD researchers to publish findings at the FLOMEKO conference.
- Built interactive dashboards for data visualization and real-time model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased inference speed by 5x using advanced AI algorithms.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute cost and time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure.
- Developed production-ready POCs using YOLO, few-shot learning, and CNNs for client demos.
- Conducted educational sessions on machine learning and big data for university students.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack website for AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and large language model frameworks.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built web scrapers and ETL pipelines for structured and unstructured data, enabling efficient data ingestion and analysis.
- Automated data cleaning workflows to support client analytics and reporting needs.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Designed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation.
- Enhanced model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time Bitcoin price forecasting model using LSTM and time-series analysis.
- Implemented data pipelines for live data ingestion and visualization of predictions.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Built the world's first AI-powered self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI in music.

Hugging Face

Open-Source

Contributor

- Co-authored Zero-shot Image Classification technical documentation and implementation page on Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Focused on machine learning, MLOps, and advanced AI systems development.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor for academic excellence.
- Specialized in data engineering, machine learning, and software development.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python